Deterministic Oracle Complexity of Nonsmooth Difference-of-Convex Optimization

Discussion with Prof. Anthony Man-Cho So

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Outline

I. **Research Goal**

II. **Preliminaries**

III. **Discussion**

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I. Research Goal

Nonsmooth DC Optimization Problem

$$F(x) = \phi(x) - g(x), \qquad \phi, g \in \Gamma_0(\mathbb{R}^d).$$

Here $\Gamma_0(\mathbb{R}^d)$ denotes proper, closed, convex functions.

- F is L-Lipschitz on $\mathbb{R}^d$
- the initialization gap is $\Delta_0 = F(x^0) - \inf_x F(x) < \infty$.
- g is nonsmooth

Research Question

How can we design a complexity-analysis framework for nonsmooth DC optimization based on a unified and computable oracle model?

Main Results

Questions

1. **Q1. What is the position, roadmap and scope of the paper? What is the real research gap in the oracle complexity of nonsmooth DC optimization?**
2. **Q2. Relations in existing approximate stationarity notions?**
3. **Q3. On the Hardness of the ε-Critical Point?**

Established

- **Q2 (VAC definition):** a value-relaxed notation for DC nonsmooth optimization.
- **Q2 (Relations to existing notions):** NAS/GAS/NAC transfer to VAC. In particular, the $\mathcal{O}(\varepsilon^{-4})$ first-order oracle complexity bound for weakly convex optimization is recovered under VAC.
- **Q3 (Definition):** The function class, exact component-wise oracle, and minimax order are specified.

Working

- Q1
- **Upper bound:** first-order oracle complexity result for nonsmooth DC optimization.
- **Lower bound:** first-order oracle complexity result lower bound for DC optimization.

II. Preliminaries

Approximate Subdifferential

Let $h : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ be proper and convex, $\delta \geq 0$, and $x \in \operatorname{dom} h$.

- **Convex δ-subdifferential:**

$$\partial_\delta h(x) := \big\{v : \ h(z) \geq h(x) + \langle v, z - x\rangle - \delta, \ \forall z\big\}, \qquad \partial_0 h(x) = \partial h(x).$$

- **Fenchel conjugate:**

$$h^*(v) := \sup_z \{\langle v, z\rangle - h(z)\}.$$

- **Fenchel–Young gap:**

$$\Phi_h(x, v) := h(x) + h^*(v) - \langle v, x\rangle, \qquad v \in \partial_\delta h(x) \iff \Phi_h(x, v) \leq \delta.$$

Remark.
δ measures the mismatch between the function value $h(x)$ and the affine dual lower model generated by s at the same point x.

DC Class and Oracle

We provide the formal definitions of the DC function class and the (component-wise) DC oracle as follows.

Definition 1 (DC Function Class)

Suppose that $\Delta, L, G > 0$. *We define* $\mathcal{D}_{\mathrm{dc}}(\Delta, L, G)$ *as the union, over dimensions* $d \in \mathbb{N}$, *of the classes of pairs*

$$(\phi, g) : \mathbb{R}^d \to \mathbb{R} \times \mathbb{R}$$

satisfying the following properties:

(i) *Both* ϕ *and* g *are proper, closed, finite-valued convex functions.*

(ii) *The DC function* $F := \phi - g$ *is* L*-Lipschitz continuous and Hadamard directionally differentiable.*

(iii) *The initialization* $\mathbf{x}^0 \in \mathbb{R}^d$ *satisfies*

$$F(\mathbf{x}^0) - \inf_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) \leq \Delta.$$

(iv) *The subgradients of both convex components are uniformly bounded:*

$$\sup_{\mathbf{x} \in \mathbb{R}^d} \sup_{\substack{\mathbf{u} \in \partial\phi(\mathbf{x}) \\ \mathbf{v} \in \partial g(\mathbf{x})}} \max\{\|\mathbf{u}\|, \|\mathbf{v}\|\} \leq G.$$

DC Class and Oracle

Definition 2 (DC Oracle)

For each query point $\mathbf{x} \in \mathbb{R}^d$, the oracle returns the values of both convex components together with their full subdifferentials:

$$\mathcal{O}_{\mathrm{dc}}(\mathbf{x}) := \big(\phi(\mathbf{x}),\, g(\mathbf{x}),\, \partial\phi(\mathbf{x}),\, \partial g(\mathbf{x})\big).$$

ε-Critical Point

Given $\varepsilon > 0$ and a pair $(\phi, g) \in \mathcal{D}(\Delta, L, G)$ defined on $\mathbb{R}^d$, a point $\mathbf{x} \in \mathbb{R}^d$ is called an ε-critical point if

$$\mathrm{dist}(\mathbf{0},\, \partial\phi(\mathbf{x}) - \partial g(\mathbf{x})) \le \varepsilon.$$

Equivalently, there exist $\mathbf{u} \in \partial\phi(\mathbf{x})$ and $\mathbf{v} \in \partial g(\mathbf{x})$ such that

$$\|\mathbf{u} - \mathbf{v}\| \le \varepsilon.$$

When $\varepsilon = 0$, $\mathbf{x}$ is called a critical point.

III. Discussion

Q1: Motivation and Research Gap

Method / Work	Problem	Stationarity	Oracle Access / Reported Cost
Weakly convex (Li–Pan '26)	Deterministic weakly convex	Moreau-envelope stationarity	$\Theta(\varepsilon^{-4})$ calls to $(F, \partial F)$
DME–GD (Sun–Sun '23)	Deterministic nonsmooth DC	Two-point ε-stationarity	$\mathcal{O}(\varepsilon^{-2})$ evaluations of $\mathrm{prox}_{\mu\phi}, \mathrm{prox}_{\mu g}$
Smooth DCA (Gratton–Toint '26)	Deterministic smooth DC:	Gradient ε-stationarity	$o(\varepsilon^{-2})$ DCA iterations;
MDCA–Polyak (Chayti–Jaggi '25)	Stochastic $H + r_1 - r_2$; smooth H	Critical distance (r_2 smooth)	$\mathcal{O}(\varepsilon^{-4})$ calls to ∇h_ξ; exact DC subproblem/iter.
SMAG (Hu et al. '24)	Stochastic DMax (incl. DWC)	Expected nearly ε-criticality (two-point)	$\mathcal{O}(\varepsilon^{-4})$ iterations; stochastic partial oracle

Sources: Li–Pan (2026), arXiv:2608.03246; Gratton–Toint (2026), arXiv:2601.15970.

SMAG access: $(\widetilde{\partial}_x \phi, \widetilde{\nabla}_y \phi, \widetilde{\partial}_x \psi, \widetilde{\nabla}_z \psi)$; the reported bound counts iterations, not deterministic $\mathcal{O}_{\mathrm{dc}}$ calls.

Research Gap

Existing results treat nonsmoothness differently and are stated under different

function class & oracle access & stationarity certificate & reported cost unit.

Hence, ε^{-2}, ε^{-3}, and ε^{-4} do not describe the same oracle-complexity problem.

Known hardness: Tian–So (2021, 2024) establish deterministic hardness for NAS/GAS, including DC-representable piecewise-linear instances. The matching component-wise DC upper bound remains open.

Q1: Core Questions

Core Questions

1. **Deterministic nonsmooth DC.** Under a component-wise deterministic DC first-order oracle and a same-point, verifiable certificate, what upper bound is achievable? Can it match $\mathcal{O}(\varepsilon^{-4})$ result?
2. **Role of smoothness.** Is the additional 2 dependence an intrinsic consequence of nonsmooth geometry, or can deterministic methods do better?

$$\underbrace{\mathcal{O}(\varepsilon^{-2})}_{\substack{\text{Sun–Sun: exact prox} \\ \text{two-point certificate}}} \quad \overset{?}{\longleftrightarrow} \quad \underbrace{o(\varepsilon^{-2})}_{\substack{\text{smooth DC} \\ \text{DCA benchmark}}}$$

Q1: Research Route and Discussion

Research Route

Unattainability $\rightarrow$ Unified DC-Aware Approximate Stationarity Certificate Design $\rightarrow$ Oracle Model $\rightarrow$ DCA Algorithms $\rightarrow$ Upper/Lower Complexity.

Discussion

1. Can the existing upper-bound analyses be unified?
2. Is the motivation sufficiently compelling?

Background

Theorem 1 (informal). There are *absolute* constants $c_0, c_1 > 0$ such that, for every finite $K > 0$, every $\varepsilon \in \big(0,\, c_0 \min\{L, G\}\big]$, and every deterministic algorithm $\mathcal{A}$ interacting with $\mathcal{O}_{\mathrm{dc}}$, there exists $(\phi, g) \in \mathcal{D}_{dc}(\Delta, L, G)$ with (Δ, L, G) independent of K such that

$$\operatorname{dist}\big(0, \partial\phi(x^k) - \partial g(x^k)\big) > \varepsilon, \qquad k = 1, \ldots, K.$$

Moreover, the critical point the pair contains $z^\star$ with $0 \in \partial\phi(z^\star) - \partial g(z^\star)$ but $0 \notin \partial_C(\phi - g)(z^\star)$.

Q2. VAC and Existing Stationarity Notions

We compare three existing finite-time relaxation notations:

1. **Near-Approximate Stationarity (NAS): Moreau relaxation.**
2. **Goldstein Approximate Stationarity (GAS): spatial convex relaxation.**
3. **Nearly Approximate Criticality (NAC): spatial DC relaxation.**

Core Result.

- Nonsmooth Nonconvex Seting:

$$\mathrm{Clarke}_{\varepsilon} \subseteq \mathrm{NAS}_{\varepsilon,\delta} \subseteq \mathrm{GAS}_{\varepsilon,\delta} \subseteq \mathrm{VAC}_{\varepsilon,\, 2G_{\phi}\delta,\, 2G_{g}\delta},$$

- Weakly Convex Setting

$$\mathrm{NAS}_{\varepsilon} \subsetneq \mathrm{VAC}_{0,\, L\varepsilon/(2\rho),\, 0}, \qquad \mathrm{VAC}_{0,\, \delta_{\phi},\, 0} \subsetneq \mathrm{NAS}_{\sqrt{8\rho\delta_{\phi}}}.$$

Ours: $(\widehat{\varepsilon}, \delta_\phi, \delta_g)$-Value Approximate Criticality (VAC)

Definition. A point x is a $(\widehat{\varepsilon}, \delta_\phi, \delta_g)$-**VAC point** if there exist u, v such that

$$u \in \partial_{\delta_\phi} \phi(x), \qquad v \in \partial_{\delta_g} g(x), \qquad \|u - v\| \leq \widehat{\varepsilon}.$$

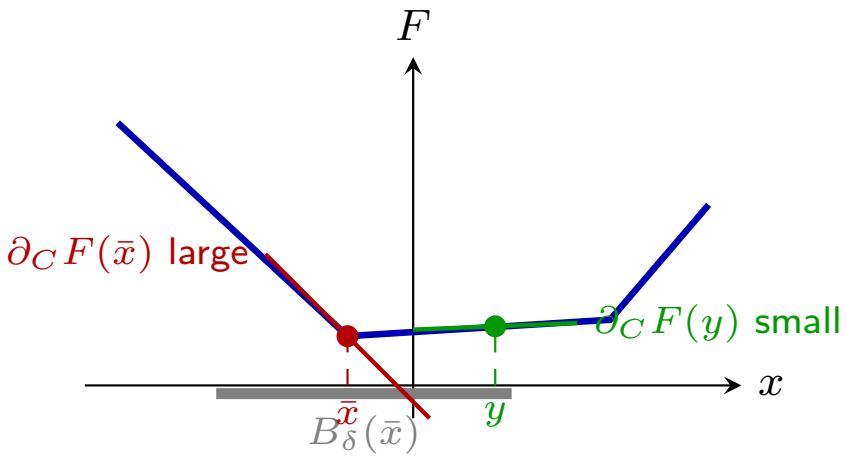
Here $\widehat{\varepsilon}$ is the vector mismatch and δ_ϕ, δ_g are the Fenchel-gap tolerance. Moreover,

$$x \text{ is } (0,0,0)\text{-critical} \iff 0 \in \partial\phi(x) - \partial g(x).$$

Spatial Relaxations (GAS) vs. Value Relaxations (VAC)

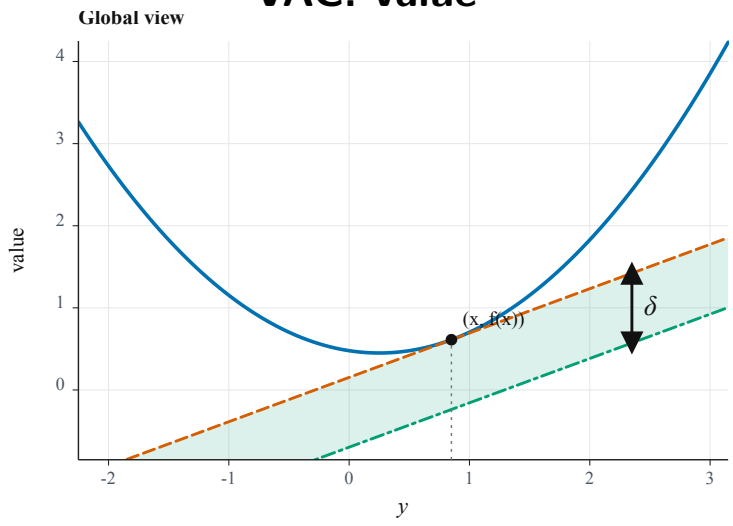
Spatial certificate vs. same-point value certificate

GAS / spatial NAS: spatial



Move to a nearby $y \in B_\delta(\bar{x})$ where the subgradient is small.

VAC: value



Stay at $\bar{x}$; shift the support by a Fenchel gap.

GAS and spatial NAS certify a neighborhood point; VAC keeps the reported point fixed.

Nearly Approximate Stationarity (NAS)

Let F be a ρ-weakly convex function with $\rho > 0$ and define

$$F_{1/(2\rho)}(x) := \inf_y \{F(y) + \rho\|y - x\|^2\}.$$

Definition (NAS)[1]

We call x an ε-NAS point of F if

$$\left\|\nabla F_{1/(2\rho)}(x)\right\| \leq \varepsilon.$$

Equivalently, if $\hat{x} = \operatorname{prox}_{F/(2\rho)}(x)$, then

$$\|x - \hat{x}\| \leq \frac{\varepsilon}{2\rho}, \qquad \operatorname{dist}(0, \partial F(\hat{x})) \leq \varepsilon.$$

One DC Decomposition of F

$$F = \phi - g, \qquad \phi(x) = F(x) + \rho\|x\|^2, \qquad g(x) = \rho\|x\|^2.$$

Here ϕ is convex and g is 2ρ-smooth convex. Moreover,

$$x \text{ is } (0, \delta_\phi, 0)\text{-VAC} \iff 2\rho x \in \partial_{\delta_\phi}\phi(x).$$

[1] Damek Davis and Dmitriy Drusvyatskiy, "Stochastic Model-Based Minimization of Weakly Convex Functions," *SIAM J. Optim.*, vol. 29, no. 1, pp. 207–239, 2019.

Proposition 1: VAC–NAS Relations

Proposition 1

(i) NAS $\Longrightarrow$ VAC. Assume that F is L-Lipschitz. For every $\varepsilon > 0$,

$$x \text{ is an } \varepsilon\text{-NAS point} \quad \Longrightarrow \quad x \text{ is a } \left(0, \frac{L\varepsilon}{2\rho}, 0\right)\text{-VAC point.}$$

(ii) VAC $\Longrightarrow$ NAS. Under the canonical decomposition

$$\phi = F + \rho\|\cdot\|^2, \qquad g = \rho\|\cdot\|^2,$$

for every $\delta_\phi \geq 0$,

$$x \text{ is a } (0, \delta_\phi, 0)\text{-VAC point} \quad \Longrightarrow \quad x \text{ is a } \sqrt{8\rho\,\delta_\phi}\text{-NAS point.}$$

In particular,

$$\delta_\phi \leq \frac{\varepsilon^2}{8\rho} \quad \Longrightarrow \quad x \text{ is an } \varepsilon\text{-NAS point.}$$

Main Finding and Discussion

Finding

- Weakly convex recovery: g is smooth, while ϕ may be nonsmooth.
- Since g is smooth,
$$\partial_C F(x) = \partial\phi(x) - \nabla g(x).$$
- A ϕ-subgradient at the output point therefore must exist; otherwise the target may be unreachable.

Discussion

- Given the nonuniqueness of DC decompositions, is this weakly convex recovery sufficient to reflect the intrinsic DC structure, or does it only describe the selected decomposition and algorithm?
- When specialized to weakly convex optimization, the DC decomposition violates the globally bounded component-subgradient assumption in our DC oracle definition; how should the function class and oracle model be reformulated?

Nearly Approximate Criticality (NAC)

We say x is an (ε, δ)-NAS point of f if

$$\operatorname{dist}\left(0, \bigcup_{y \in \mathbb{B}_\delta(x)} \partial f(y)\right) \leq \varepsilon.$$

Definition (Sun–Sun '23): $(\widehat{\varepsilon}, \delta)$-NAC[2]

Given $\widehat{\varepsilon}, \delta \geq 0$, we call x a $(\widehat{\varepsilon}, \delta)$-NAC point for $F = \phi - g$ if

$$\operatorname{dist}\left(0, \partial\phi(x) - \bigcup_{y \in B_\delta(x)} \partial g(y)\right) \leq \widehat{\varepsilon}.$$

- $\widehat{\varepsilon}$: dual-vector mismatch;
- δ: spatial displacement between x and y.

[2]Kaizhao Sun and Xu Andy Sun, "Algorithms for Difference-of-Convex Programs Based on Difference-of-Moreau-Envelopes Smoothing," *INFORMS Journal on Optimization*, vol. 5, no. 4, pp. 321–339, 2023.

NAC to VAC

Proposition: NAC to VAC

Assume that g is (locally) G-Lipschitz. Then

$$(\widehat{\varepsilon}, \delta) - \text{NAC} \Longrightarrow (\widehat{\varepsilon}, 0, 2G\delta)\text{-VAC}.$$

- Note that the converse fails in general.
- Under the weakly convex split,

$$(\widehat{\varepsilon}, \delta)\text{-NAC} \quad \Longleftrightarrow \quad \operatorname{dist}\big(0, \partial_C F(x)\big) \leq \widehat{\varepsilon} + 2\rho\delta,$$

so it inherits the finite-time unattainability of approximate Clarke stationarity;

Goldstein Approximate Stationarity (GAS)

Definition (Goldstein '77; Burke–Lewis–Overton '05)

For a locally Lipschitz F, we say that z is a (ε, δ)-GAS point if

$$\operatorname{dist}\Big(0, \operatorname{conv}\Big(\bigcup_{y \in B_\delta(z)} \partial_C F(y)\Big)\Big) \leq \varepsilon.$$

Proposition: GAS to VAC

Assume that both convex components are (locally) G-Lipschitz. Then

$$(\varepsilon, \delta)\text{-GAS} \implies (\varepsilon, 2G\delta, 2G\delta)\text{-VAC}.$$

The implication uses $\partial_C(\phi - g)(y) \subseteq \partial\phi(y) - \partial g(y)$ and transports each component subgradient from $y \in B_\delta(z)$ to z.

$\mathcal{O}(\varepsilon^{-4})$ First-Order Complexity Recovery

Algorithm 1: Inexact DCA under $\mathcal{O}_{DC}$

- At the k-th iteration, consider the DCA subproblem

$$\min_x \; F_k(x) = \phi(x) - \langle \nabla g(x^k), x \rangle \equiv \min_x \; \left\{ F(x) + \rho \|x - x^k\|^2 \right\},$$

  which is ρ-strongly convex.

- **Inner solver:** Run J subgradient descent steps on $B(x^k, G/\rho)$ to obtain x^{k+1} satisfying

$$F_k(x^{k+1}) \leq \inf_x \{F_k(x)\} + a,$$

  where $a > 0$ is the prescribed inner accuracy.

- **Output:** Define

$$\bar{\delta}_k := F_k(x^k) - F_k(x^{k+1}) + a \geq 0.$$

  Return $x^{k_\star}$, where $k_\star \sim \mathrm{Unif}\{0, \ldots, K-1\}$ with K iteration.

$\mathcal{O}(\varepsilon^{-4})$ First-Order Complexity Recovery

Theorem

Let $F : \mathbb{R}^d \to \mathbb{R}$ be ρ-weakly convex with $\rho > 0$ and G-Lipschitz, and let $\Delta_0 := F(x^0) - \inf F \in (0, \infty)$. For any $\varepsilon > 0$, set

$$a = \frac{\varepsilon^2}{16\rho}, \qquad K = \left\lceil \frac{16\rho\Delta_0}{\varepsilon^2} \right\rceil, \qquad J = \mathcal{O}\left(\frac{G^2}{\varepsilon^2}\right).$$

Then the output $x^{k_\star}$ is a $(0, \bar{\delta}_{k_\star}, 0)$-AC point and, with expectation over $k_\star$,

$$\mathbb{E}\left[\bar{\delta}_{k_\star}\right] \le \frac{\varepsilon^2}{8\rho} \quad \Longrightarrow \quad \mathbb{E}\left[\left\|\nabla F_{1/(2\rho)}(x^{k_\star})\right\|^2\right] \le \varepsilon^2.$$

The total first-order oracle complexity is $\mathcal{O}(\rho\Delta_0 G^2/\varepsilon^4)$.

Remark

Through the AC-to-NAS bridge, Algorithm 1 recovers the standard $\mathcal{O}(\varepsilon^{-4})$ first-order oracle complexity for weakly convex near-stationarity in expectation.[3]

[3] Jiajin Li and Siyu Pan, "Optimal Deterministic Oracle Complexity for Weakly Convex Optimization," arXiv:2608.03246, 2026.

Q3. DC Oracle Complexity: Five Core Objects

1. **Problem instance.**
$$I = (\phi, g), \qquad F_I = \phi - g.$$
2. **Admissible class.** $I \in \mathcal{D}_{\mathrm{dc}}(\Delta, L, G)$, where Δ controls the initial gap, L controls F_I, and G controls component subgradients.
3. **Information oracle.** At a query x, the algorithm observes
$$\mathcal{O}_{\mathrm{dc}}(I, x) = \big(\phi(x), g(x), \partial\phi(x), \partial g(x)\big).$$
4. **Algorithm and query sequence.** $\mathcal{A} \in \mathfrak{A}_{\mathrm{det}}(\mathcal{O}_{\mathrm{dc}})$ chooses $x^k_{\mathcal{A}[I]}$ using only the preceding oracle transcript.
5. **Success criterion.**
$$R_I(x) := \mathrm{dist}(0, \partial\phi(x) - \partial g(x)) = \inf_{\substack{u \in \partial\phi(x) \\ v \in \partial g(x)}} \|u - v\|.$$
A query is successful when $R_I(x) \le \varepsilon$.

Q3. Reading the Min–Max Hardness Statement

With the five objects fixed, the deterministic finite-budget statement is

$$\underbrace{\inf_{\mathcal{A}\in\mathfrak{A}_{\mathrm{det}}(\mathcal{O}_{\mathrm{dc}})}}_{\text{best algorithm}} \quad \underbrace{\sup_{I\in\mathcal{D}_{\mathrm{dc}}(\Delta,L,G)}}_{\text{hardest admissible instance}} \quad \underbrace{\min_{0\le k<K}}_{\text{best queried point}} \quad R_I\left(x^k_{\mathcal{A}[I]}\right) > \varepsilon.$$

Read the quantifiers from inside to outside

1. $\min_{k<K}$ gives the algorithm credit for its best point among the first K queries.
2. $\sup_I$ selects the hardest admissible DC pair after $\mathcal{A}$ has been fixed.
3. $\inf_{\mathcal{A}}$ asks whether even the best deterministic algorithm can guarantee success.

Equivalently, the target theorem has the quantifier form

$$\forall \mathcal{A} \in \mathfrak{A}_{\mathrm{det}}, \quad \exists I \in \mathcal{D}_{\mathrm{dc}}(\Delta, L, G): \quad R_I\left(x^k_{\mathcal{A}[I]}\right) > \varepsilon, \quad k = 0, \ldots, K-1.$$

Current status

The oracle model and minimax statement are fixed. A DC hard instance proving this statement with budget-independent (Δ, L, G) is still missing.

[4]

[4] Lai Tian and Anthony Man-Cho So, "No Dimension-Free Deterministic Algorithm Computes Approximate Stationarities of Lipschitzians," *Mathematical Programming*, vol. 208, nos. 1–2, pp. 51–74, 2024, Theorem 1.

Summary and Future Work

Summary and Future Work

What is established in this discussion

- VAC is a same-point, decomposition-dependent Fenchel-gap certificate; zero tolerances recover exact DC criticality.
- Under the canonical quadratic split, VAC and NAS are quantitatively linked, but not equivalent at the same tolerance.
- VAC differs from Clarke/Goldstein stationarity: it relaxes values at one point rather than sampling a neighborhood.
- Under an F-first-order oracle, inexact DCA gives an expected $\mathcal{O}(\varepsilon^{-4})$ NAS upper bound.

What remains open

- Q3: finite-time attainability of exact DC criticality under the component-wise DC oracle.
- Q3: a hard-instance construction with oracle indistinguishability and a DC-residual reduction.
- Q3: a common function class and oracle model for a meaningful DC upper/lower-complexity comparison.

Takeaway

VAC gives a finite-time same-point DC certificate; its NAS bridge is explicit, while a matching DC-oracle lower bound remains open.